

Introduction of the project

Water is an essential resource used in daily life for domestic, agricultural, and industrial purposes. Proper management of water is important to avoid wastage and ensure efficient usage. In many households and buildings, water is stored in overhead tanks, but monitoring the water level manually can be inconvenient and sometimes inaccurate.

Due to the lack of proper monitoring, problems such as overflow of water tanks and dry running of pumps are very common. Overflow leads to unnecessary wastage of water, while dry running can damage the pump and reduce its efficiency. These issues create the need for a simple and reliable system to monitor water levels.

The Water Level Indicator is an electronic system designed to indicate the level of water in a tank using probes placed at different heights. It provides visual indications using LEDs and can also give audio alerts using a buzzer when the tank is full or empty.

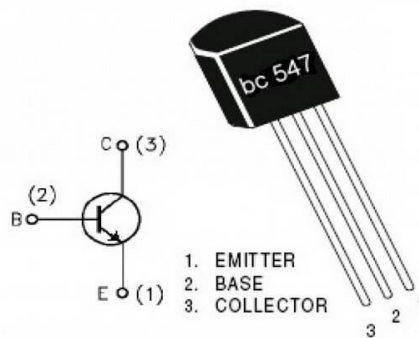
The working of this project is based on the conductivity of water. When water touches the probes, it allows a small current to flow, which activates transistors in the circuit. These transistors then turn ON the respective LEDs or buzzer, indicating the water level.

This project is simple, cost-effective, and easy to implement using basic electronic components. It helps in conserving water, reducing manual effort, and preventing damage to equipment. It also serves as a good practical example for understanding basic electronic concepts.

Introduction of each component

1. BC547 Transistor

The BC547 is an NPN bipolar junction transistor used as a switch in the circuit. When a small current is applied at its base, it allows a larger current to flow from collector to emitter. This switching action helps in turning ON LEDs or buzzers when water reaches a certain level.



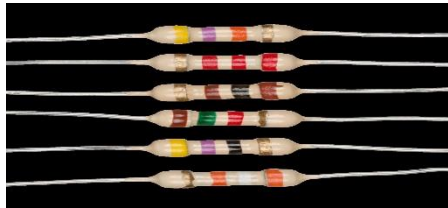
2. Probes (Sensors)

Probes are simple metal wires or rods placed at different heights inside the water tank. They act as sensors. When water touches both the common probe and any level probe, it completes the circuit and allows current to flow.



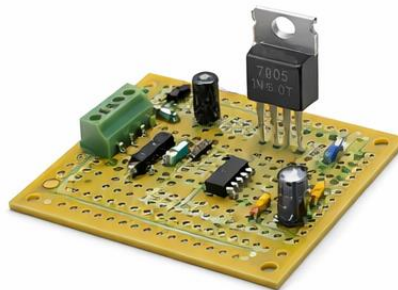
3. Resistors

Resistors are used to control and limit the flow of current in the circuit. They protect components like transistors and LEDs from damage due to excessive current and also help in controlling LED brightness.



4. PCB (Printed Circuit Board)

The PCB is used to mount and interconnect all the electronic components in a compact and organized manner. It replaces loose wiring and ensures proper electrical connections through copper tracks. Using a PCB improves the durability, reliability, and neatness of the circuit.



PCB (Printed Circuit Board)

5. USB Cable / Output Port

LEDs provide visual indication of water levels:

- Green LED → Low level
- Yellow LED → Medium level
- Red LED → Full level

They glow when the corresponding water level is reached.



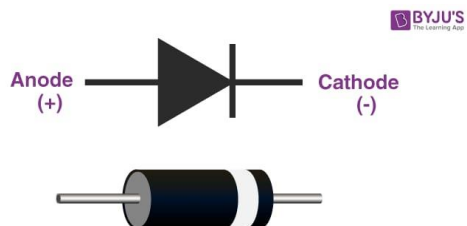
6. Buzzer

The buzzer provides an audio alert. It is used to indicate important conditions such as when the tank is full or when the water level is very low.

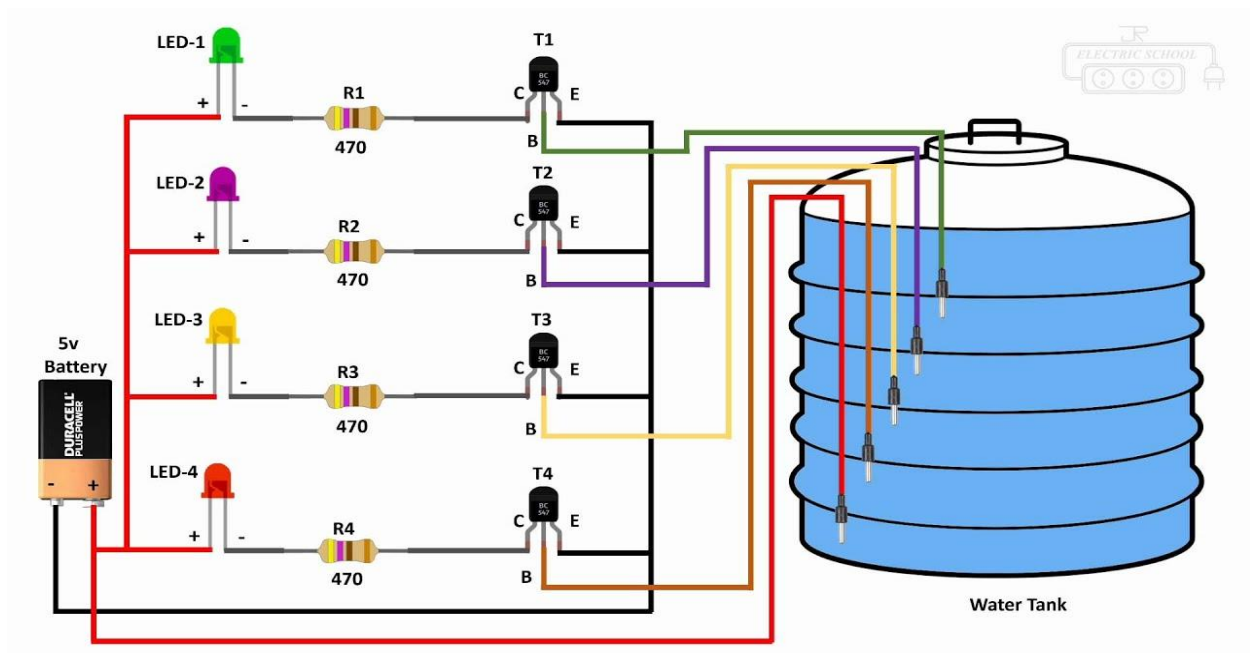


7. Diode

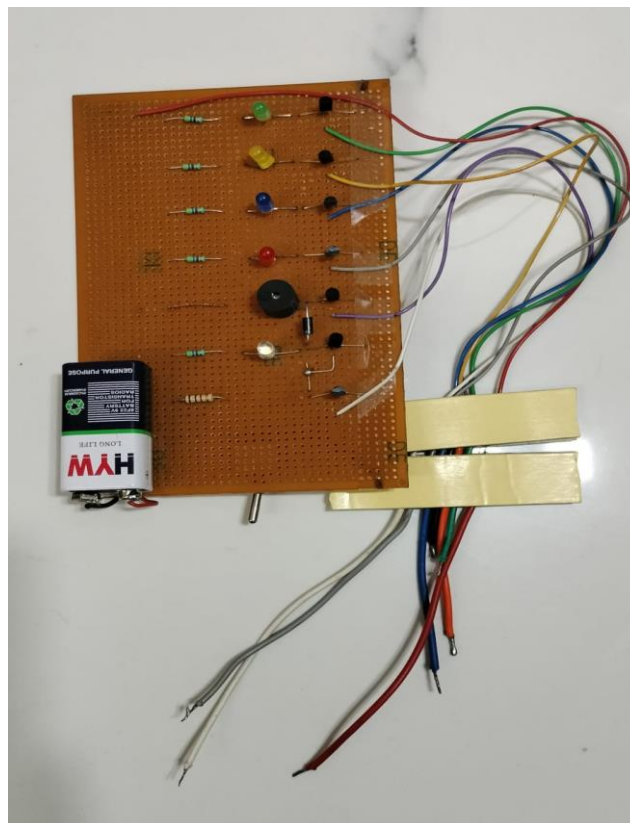
A diode is used to allow current to flow in only one direction. In this circuit, it prevents reverse current flow from the output side back to the solar panel, especially when there is no sunlight. This helps in protecting the components and improving efficiency.



Circuit Diagram



PCB Circuit Constructed image



Principle

The Water Level Indicator works on the basic principle of electrical conductivity of water and the switching action of transistors. Most natural water contains dissolved salts and minerals, which make it a good conductor of electricity. This property is used in the circuit to detect the presence and level of water in a tank.

In this system, a set of metal probes is placed at different heights inside the water tank. One probe acts as a common reference (usually placed at the bottom), while the others represent different water levels such as low, medium, and full. When the water level rises and touches any of these probes along with the common probe, it completes an electrical path between them.

As soon as this path is formed, a small current flows through the water from the probe to the base of a transistor (BC547). Even though the current is very small, it is sufficient to activate the transistor. The transistor then operates in switching mode, turning ON and allowing a larger current to flow from collector to emitter.

This switching action is used to drive output devices such as LEDs and buzzers. Each probe is connected to a separate transistor circuit, so when water reaches a particular level, the corresponding transistor turns ON and activates its indicator. In this way, the circuit provides different indications for different water levels.

Resistors are used in the circuit to limit the current and protect the components from damage. A diode may also be included to ensure that current flows only in one direction and to protect the circuit from reverse voltage.

Thus, the system converts the physical level of water into electrical signals using conductivity and then into visual and audio outputs using electronic components. This makes it a simple yet effective method for monitoring water levels without the need for complex sensors or programming.

Working

When the circuit is powered using a 9V or 12V DC supply, it becomes active.

A common probe is placed at the bottom of the tank, and other probes are placed at different levels (low, mid, full).

- **No Water Condition:**
When water is below the low-level probe, no current flows. The low-level LED may glow as a power indicator.
- **Low-Level Detection:**
When water touches the low-level probe and common probe, current flows through water. The transistor turns ON, and the low-level LED glows.
- **Mid-Level Detection:**
As water rises further, it touches the mid-level probe. The second transistor activates, and the mid-level LED glows.
- **Full-Level Detection:**
When water reaches the top probe, the third transistor turns ON. The full-level LED glows, and the buzzer may also activate.
- **Water Level Falling:**
As water decreases, the full LED turns OFF first, then mid-level, and finally low-level indication disappears.

Output

The output of the water level indicator is obtained in the form of visual and audio signals.

As the water level in the tank changes, the corresponding LEDs glow to indicate different levels such as low, medium, and full. When the water reaches a particular probe, the respective LED turns ON, providing a clear visual indication of the water level. In addition to this, a buzzer may also be activated at critical levels, such as when the tank is completely full or when the water level is very low, giving an audible alert to the user.

This real-time indication helps in monitoring the water level without manual checking. The system ensures that users can easily understand the status of the tank and take necessary actions, such as turning the pump ON or OFF, thereby preventing water wastage and protecting the pump from damage.

Applications

- Used in home water tanks to know the exact water level.
- Helps prevent tank overflow by giving a full-level alert.
- Prevents pump dry running when water is too low.
- Useful in apartments, societies, and hostels for common water tanks.
- Can be used in industries to monitor liquid levels in storage tanks.
- Helpful in agriculture for monitoring irrigation water tanks.
- Can be combined with a relay to make a fully automatic water pump controller.
- Helps in water conservation by avoiding unnecessary wastage.

Limitations of the Project

- Accuracy depends on water conductivity
- Probes may corrode over time
- Not suitable for very pure (distilled) water
- Requires proper placement of probes
- Limited to small-scale applications
- No automatic control unless extended

Suggestions / Modifications

- Use stainless steel probes to reduce corrosion
- Add a relay to automate pump ON/OFF
- Include a display (LCD) for precise level reading
- Use a microcontroller (like Arduino) for advanced control
- Add wireless monitoring system
- Use waterproof sensors for better durability